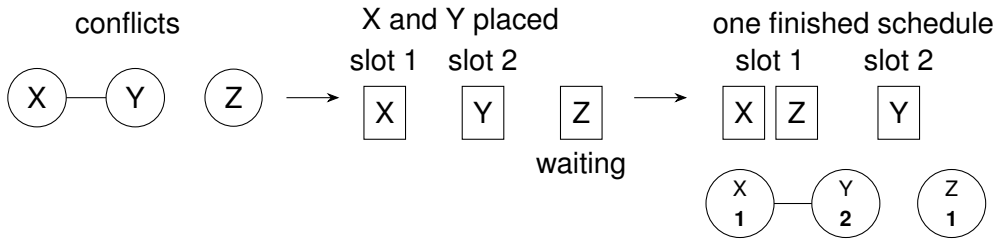
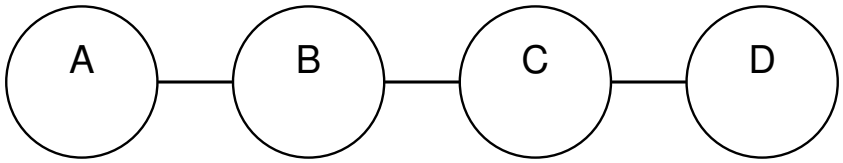
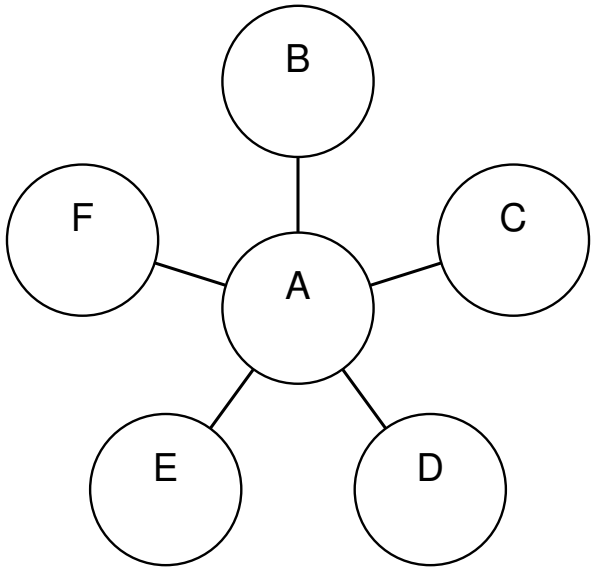
Each circle is one activity card. A line means its two cards cannot share a slot. Put each card in exactly one slot. A slot can hold any number of cards if none of them conflict. Only marked circles are activities; line crossings add none. Write a card's slot number in its circle.




**Problem 1:** Use your cards to schedule each network with the fewest slots you can.

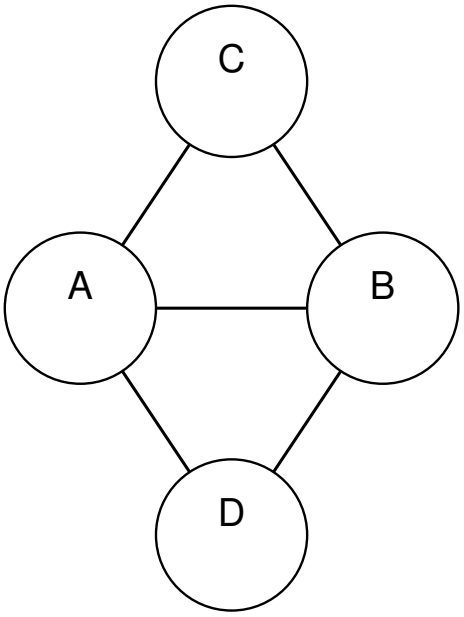



Fewest slots: _____

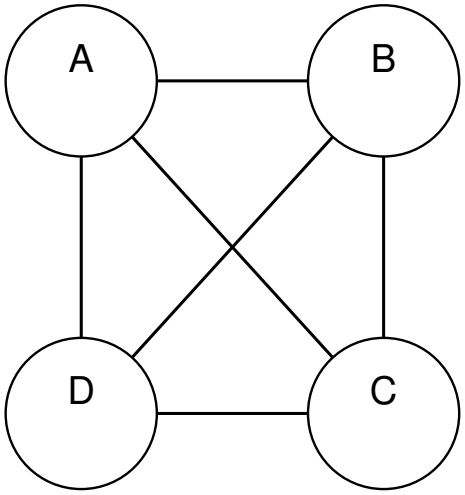



Fewest slots: _____

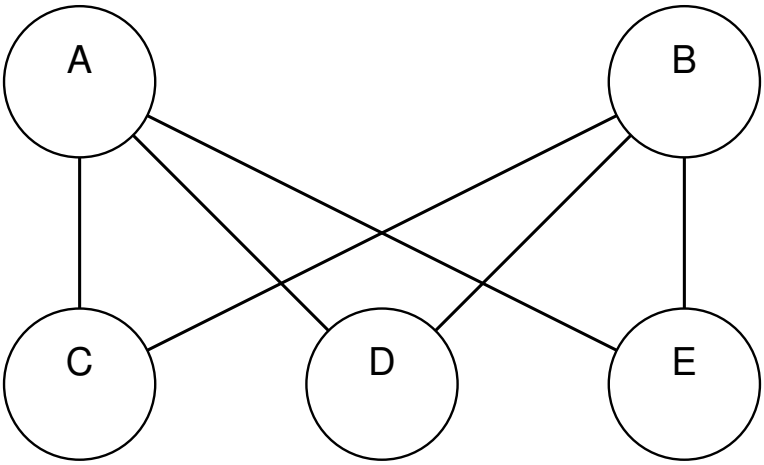
**Problem 2:** Find the fewest slots for each network. Show why fewer slots cannot work.




Fewest slots: _____

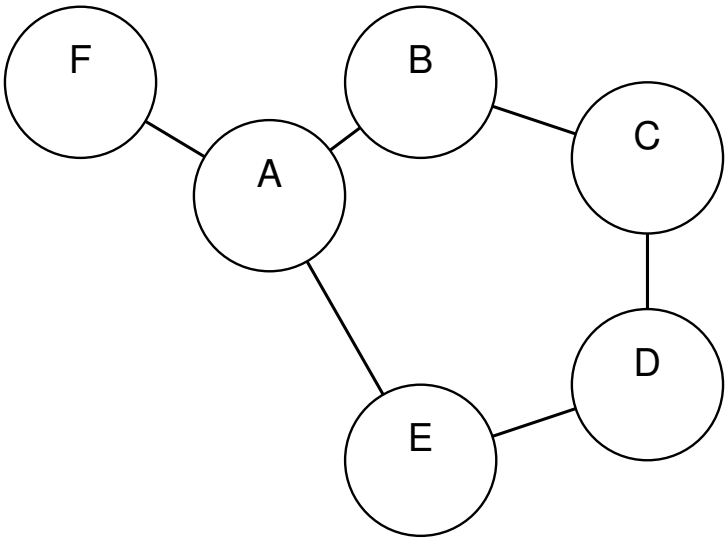



Fewest slots: _____

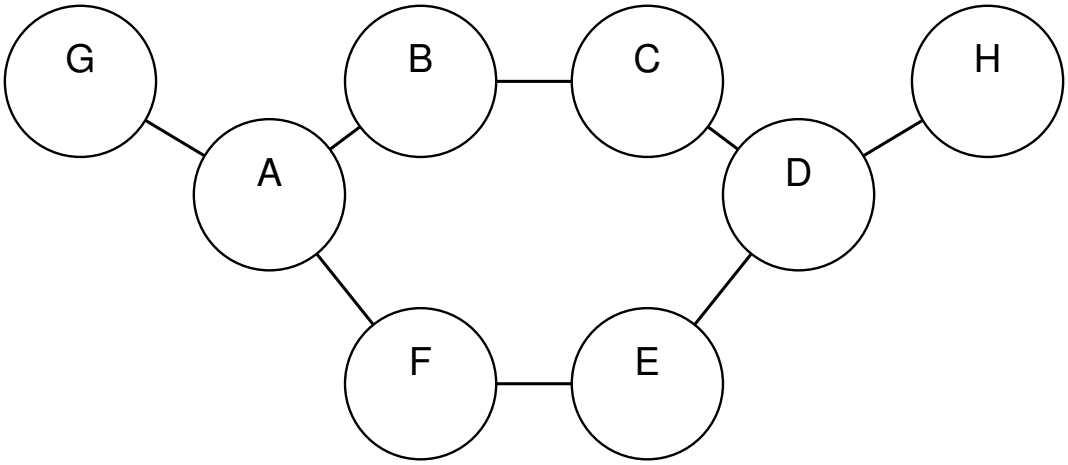



Fewest slots: _____

**Problem 3:** Find the fewest slots for each network. Also find the largest group of cards in which every pair conflicts. Does that group's size always give the fewest slots?

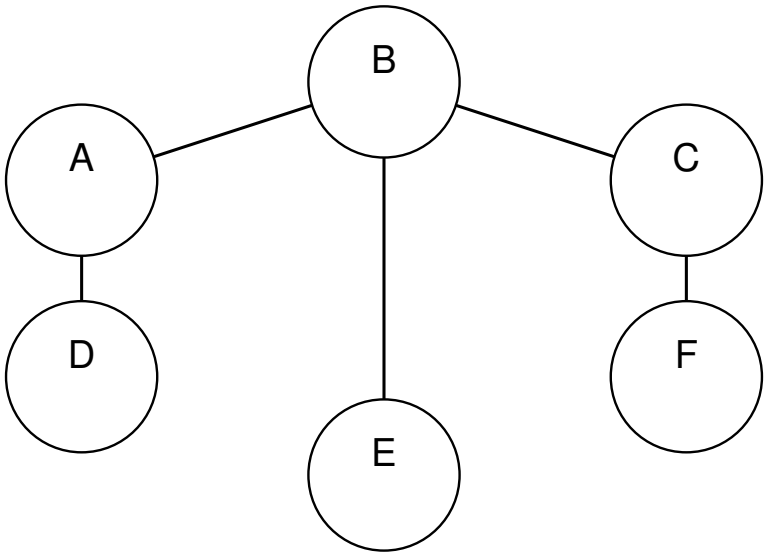



Fewest slots: _____ Largest group: _____

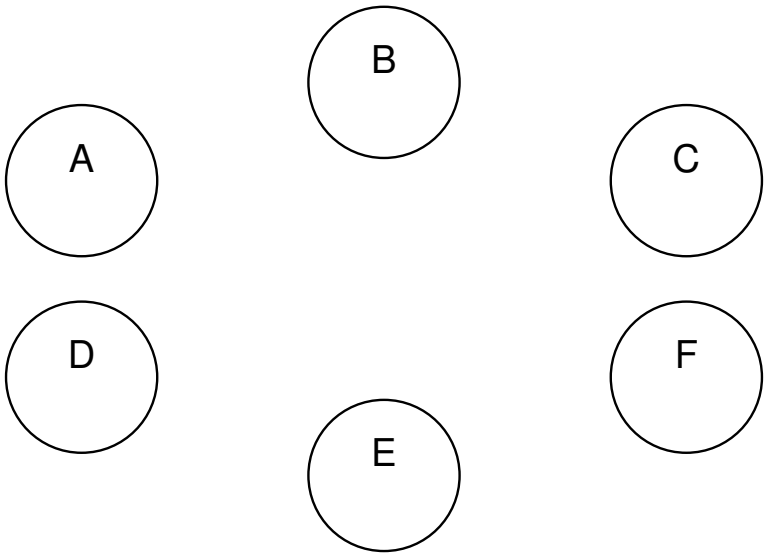



Fewest slots: _____ Largest group: _____

**Problem 6:** Add the fewest conflict lines you can so each network can no longer fit in two slots. Join only different circles that have no line yet. Show why fewer added lines cannot work.

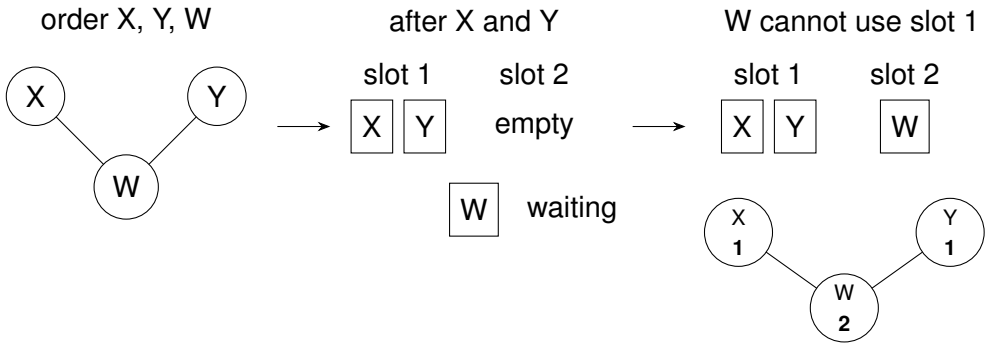



Added lines: _____

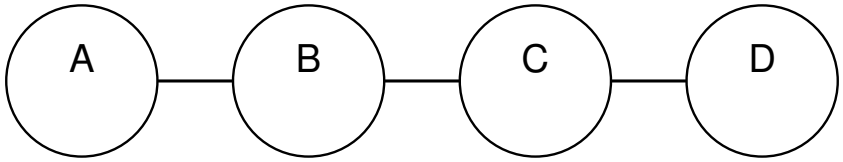



Added lines: _____

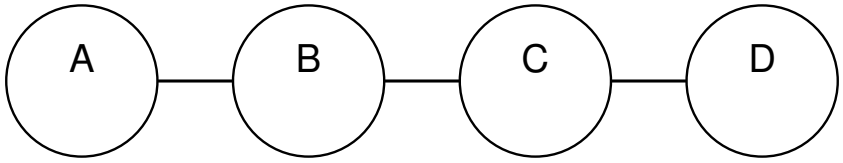
For Problems 7–8, choose an order for the cards. Place each card in the first numbered slot where it conflicts with no card already there. During the order, do not move a placed card.




**Problem 7:** Find orders that make the first-slot rule use as few slots and as many slots as possible on this network. After an order is finished, you may move cards. Can you use fewer slots?




Order: _____




Order: _____